These differences illustrate that there's no single *right* configuration. Each design must fit the risk, scale, and assurance needs of the system it serves.

Normalization transformed these point solutions into a unified enterprise capability. With a shared PIP and consistent schema, ACME eliminated duplicate integrations, unified signal transmission, and sped up the adoption of fine-grained authorization across the organization. Every PDP now receives the same high-quality context, regardless of which system makes the request.

Not every signal follows the same path, however. Some—like device posture or short-lived risk attestations—arrive as tokens with the request, while others flow continuously through the enterprise PIP. In ACME's hybrid model, tokens carry only the minimal identifiers and scopes needed for verification, while the PIP supplies the richer background attributes that policies depend on. This design keeps tokens lightweight, limits exposure, and maintains consistency across all decision points.

In effect, ACME's identity team turned authorization context into its own platform, an internal service that delivers reliable, verifiable signals at every decision point. Policies have become portable, auditable, and easier to understand because they depend on a shared foundation of standardized data.

As the next chapter shows, this contextual layer enables effective use of policy engines within existing systems. The focus shifts from collecting and validating signals to integrating authorization decisions into the systems and workflows that rely on them.

12.8 Summary

- Policies are only as good as their signals; the three design dimensions are what signals are used, how they are delivered, and how fresh they are.
- Real-world PIPs use signals from multiple sources. PDPs feature lightweight processing before the policy engine.
- Token, query, and hybrid delivery each trade off latency, availability, and assurance.